
WINTER PROJECT 2018

SkyHopper Exoplanet Detection:

Transit Probability Simulation

Yi Shuen Lee (Christine)

June 25 - July 20, 2018



Institution: School of Physics,
The University of Melbourne
Supervisor: Professor Michele Trenti

Contents

1	INTRODUCTION	3
1.1	SkyHopper and its Vision	3
1.2	Project Overview	3
2	PLANETARY TRANSIT	4
3	SIMULATION INPUT	5
3.1	SkyHopper Visibility	5
3.2	Variables and Derived Quantities	6
4	SIMULATION PROCEDURES	7
4.1	Defining Planetary Transit Function	7
4.2	Random Number Generators	9
4.2.1	Observation Start Time	9
4.2.2	Planet's Location in Orbit	9
5	SIMULATION RESULTS	10
5.1	Potential Exoplanet Transit Detection	10
5.2	Detailed Visualisation of Potential Transit Detection	12
6	TRANSIT DETECTION EFFICIENCY	14
6.1	Exoplanet Transit Detection Condition	14
6.2	Transit Detection Efficiency	14
7	SIMULATION OUTPUT	16
7.1	Sample Monte Carlo Simulation	16
7.2	Output Plots for Individual Run	16
7.3	General Information Output	17
7.4	Information for Individual Runs	17
7.5	Transit Steps Detected per Orbit	19
8	SUMMARY OF SIMULATION (FLOWCHART)	20
9	WHAT'S NEXT?	21

1 INTRODUCTION

1.1 SkyHopper and its Vision

The SkyHopper is a 12U, near-infrared (NIR) cubesat designed with a major objective to identify the earliest Gamma Ray Burst (GRB) afterglows at the furthest observable surface of the universe. However, GRB afterglows cannot be observed all the time, so the part of the remaining time of SkyHopper will be used to:

- a. search for earth-like planets
- b. explore the time-variable universe (e.g. identifying transient phenomena)
- c. measure the Cosmic Microwave Background (CMB) radiation
- d. unveil planetary system formation

More information on SkyHopper can be found on the official [website](#)^[1].

1.2 Project Overview

This winter project will focus on the objective of searching for exoplanets around low mass stars using SkyHopper. For most of SkyHopper's functional time, it is not actually looking for GRBs since the observation procedure is highly condition dependent. So in this spare time, we are planning to aim SkyHopper at a selected patch of the sky, hoping to detect as many planetary transit as it can during this time.

This project aims to perform Monte Carlo simulations to determine the probability of exoplanet transit detection by SkyHopper. However, SkyHopper cannot observe a fixed patch of the sky continuously as the Earth comes in the way during SkyHopper's orbit around the Earth. So the general idea is here to convolve the transit light curve of a star with the visibility map of SkyHopper. This will tell us at which visible timesteps would SkyHopper detect planetary transits. The all-sky visibility map of SkyHopper is provided by Robert Mearns.

2 PLANETARY TRANSIT

There are many different techniques to find exoplanets. These include:

- Radial velocity
- Transit (most successful)
- Direct imaging
- Gravitational microlensing
- Astrometric wobble

The method used for exoplanet detection here, as mentioned in section 1.2, is the transit method, which only applies to planets with orbital plane approximately parallel to our line of sight. For simplicity in this project, we made the assumption that all the planets we observe satisfy this condition.

As a planet passes in front of a star during its orbit, it partially blocks the light of the star from reaching the observer, causing a significant reduction in the flux measured by an observer. This dip in the light curve is known as a planetary transit.

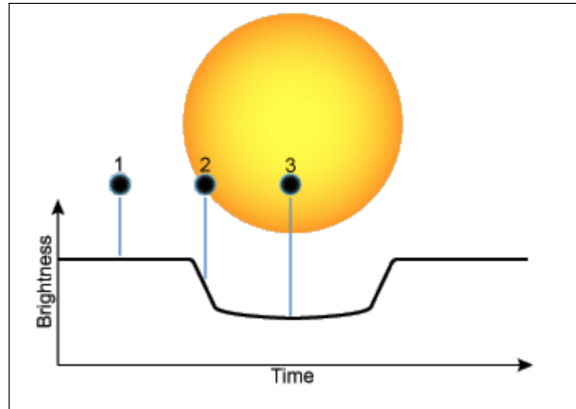


Figure 2.1: Light curve of a star during planetary transit. (1, 2 and 3 indicates position of the planet with respect to the point on the light curve.)

The depth of the dip in the light curve can provide information on the size of the planet. The greater the depth of the light curve, the larger the size of the planet^[2]. Also, by analogy to the orbital period of, say, Jupiter and Earth, we know that the further away the planet is from the star, the longer its orbital period would be. Hence the longer the duration of light curve dip, the further out the planet is from its host star.

However, we have to be very careful in identifying these light curve dips as they could possibly be false signals caused by instrumental errors or some other non-planetary bodies passing by the front of the host star.

3 SIMULATION INPUT

3.1 SkyHopper Visibility

SkyHopper cannot continuously stare at the same patch of the sky. This is because for about half of its orbital time (~ 96 min), its vision on that specific patch of the sky is occulted by the Earth itself. This has to be taken into account when generating the simulation of exoplanet transit detection.

The SkyHopper visibility map provides information on the visibility of SkyHopper on every location across the sky at any time of the year. The raw data comes as a 3-dimensional array: [timestep, Ra, Dec], with each timestep representing a 6 minute cadence whereas Ra and Dec means Right Ascension and Declination, which gives the location in the sky. On the visibility axis, the value "1" indicates that the selected [Ra,Dec] is visible during that timestep of the year, whereas value "0" indicates that it is not visible.

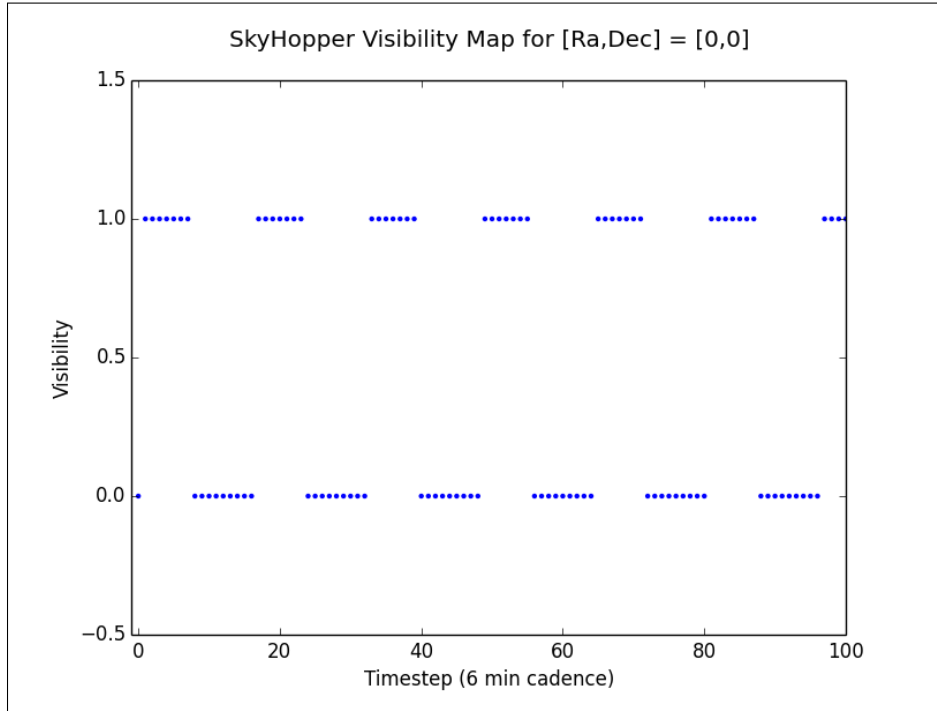


Figure 3.1: SkyHopper visibility map of [Ra,Dec]=[0,0] for the first 100 timesteps (i.e. first 600 minutes) of the year where each plot point represents one discrete 6-minute timestep.

3.2 Variables and Derived Quantities

This simplified simulation will explore two varying parameters namely the *stellar mass*, M , and *orbital period*, T . Given the input of these two values, many other important quantities required for the simulation can be derived. These *derived quantities* include^[3]:

a. **Stellar Radius, R :**

Given the stellar mass-radius relationship

$$R = kM^x \quad (1)$$

We assume that all the stars within the parameter space are main sequence stars hence we take the values $k = 1$ and $x = 0.8$. Hence the stellar radius calculation goes like:

$$R = M^{0.8} \left(\frac{R_{\odot}}{M_{\odot}^{0.8}} \right) \quad (2)$$

b. **Semi-major axis, a :**

By Kepler's Third Law of Planetary Motion

$$a = \left(\frac{T^2 G (M + m)}{4\pi^2} \right)^{\frac{1}{3}} \quad (3)$$

where m = mass of planet. However, since $m \ll M$, we can assume that

$$a \approx \left(\frac{T^2 G M}{4\pi^2} \right)^{\frac{1}{3}} \quad (4)$$

c. **Planetary Transit Duration, t :**

$$t = 13 \left(\frac{M}{M_{\odot}} \right)^{-\frac{1}{2}} \left(\frac{a}{1AU} \right)^{\frac{1}{2}} \left(\frac{R}{R_{\odot}} \right) \text{ hours} \quad (5)$$

Constants:

$M_{\odot}$ = Solar mass = $2 \times 10^{30} kg$

$R_{\odot}$ = Solar radius = $6.957 \times 10^8 m$

G = Gravitational Constant $6.67 \times 10^{-11} m^3 kg^{-1} s^{-2}$

4 SIMULATION PROCEDURES

4.1 Defining Planetary Transit Function

With the planetary transit duration derived for a given M and T , we can use it to define the light curve (luminosity-time) of the host star. Again for simplicity, we are not taking into account the gradual flux change during a planetary transit as shown in point "2" of Figure 2.1. We are assuming that there is a direct drop in luminosity as the start of a transit.

In this simulation, the start of a new planetary orbit is at the instant where a transit starts, and is defined by planet orbital location timestep, $n_{planet} = 0$. To make that clearer, the planet orbit location timestep at 6 minutes after the start of a transit would be $n_{planet} = 1$ as 6 minutes is equivalent to 1 timestep.

The number of timesteps in one full planetary orbit, N_{planet} , is:

$$N_{planet} = \frac{T}{6} \quad (6)$$

Hence, the location of the planet, x_{planet} , in orbit can be defined by:

$$x_{planet} = \frac{n_{planet}}{N_{planet}} \quad (7)$$

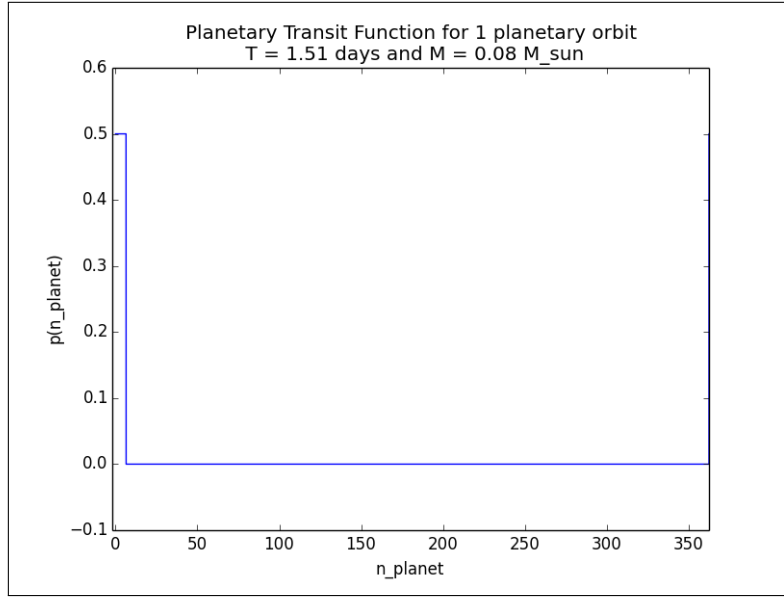
However, the "light curve" used in this simulation is slightly different from the one shown in Figure 2.1. What is used here should be called a "Planetary Transit Function", $p(n_{planet})$ rather than a light curve which should, technically speaking, be a luminosity vs. time plot. So $p(n_{planet})$ is essentially a piecewise continuous step function defined as such:

$$p(n_{planet}) = \begin{cases} 0.5 & n_{planet} \leq n_{transit} \\ 0 & n_{transit} < n_{planet} \leq N_{planet} \end{cases} \quad (8)$$

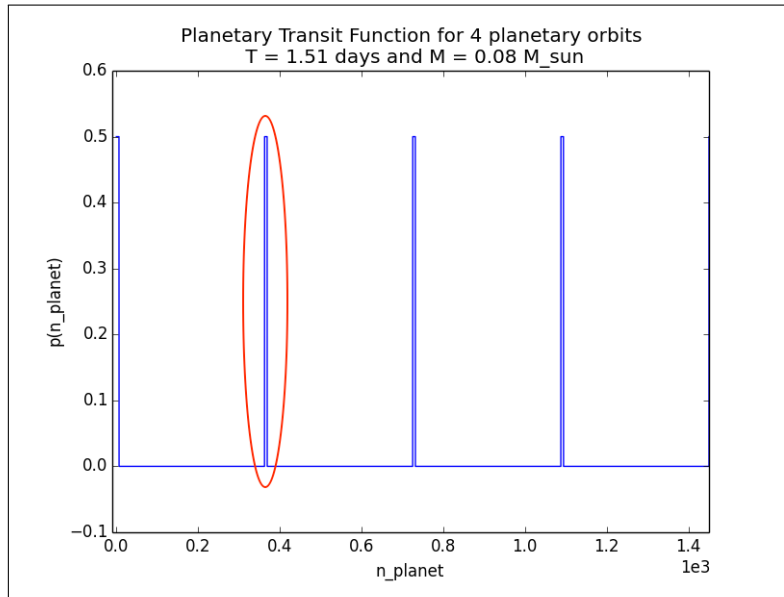
In words, $p(n_{planet}) = 0.5$ when the planet is in transit and $p(n_{planet}) = 0$ when the planet is not in transit.

This piecewise function is analogous to the light curve of one full planetary orbit as it provides the exact same information but just represented inversely. So instead of a dip, the planetary transit is represented as a bump/spike.

However, since $p(n_{planet})$ only represents one orbit (Figure 4.1a), we have to take modulo N_{planet} of the array of selected orbital timestep to generate periodic Planet Transit Functions (Figure 4.1b).



(a) Planetary Transit Function for 1 full planetary orbit.



(b) Planetary Transit Function for 4 full planetary orbits.

Figure 4.1: The spike(s) in both figures represent the transit. (One of the spikes is circled in red for illustration.)

4.2 Random Number Generators

4.2.1 Observation Start Time

The script has a random number generator that will produce a random timestep during the year at which SkyHopper will start its observation on a specific patch of the sky.

However, there could be, for example, a possible circumstance that the random start timestep is in the last week of the year while the set observation period is more than a week. This means that the observation extends across a new year. In such cases, the code will reset the visibility map and proceed to use the visibility map at the start of a new year.

4.2.2 Planet's Location in Orbit

Another random variable to consider is the location of the planet in its orbit at the start of the observation. In other words, we don't know where the planet will be in its orbit at the start of the observation.

To generate this random location, the random number generator is set to produce a random value, n_{start} between 0 and N_{planet} , which would then also give a random start location in the orbit, x_{start} . This would mean the planet may or may not be in transit at the start of the location.

We can then again use the modulo of N_{planet} as per Section 4.1 to produce a periodic Planet Transit Function that stretches over the observation period.

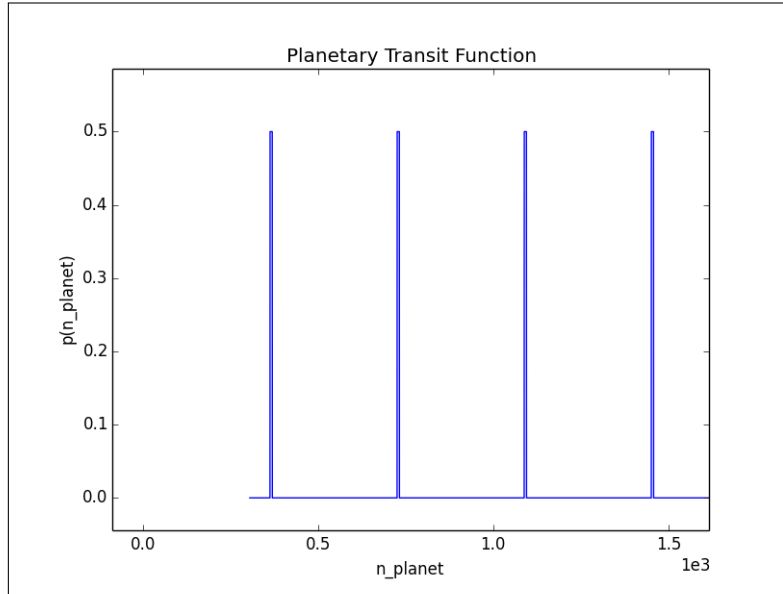


Figure 4.2: Example of a random orbit start. Notice that the start of the Planetary Transit Function is not at $n_{planet} = 0$.

5 SIMULATION RESULTS

5.1 Potential Exoplanet Transit Detection

With the visibility map of SkyHopper and the Planetary Transit Function over the observation period, we can now, in a sense, convolve the two functions. Convolution here is simple. It is just multiplying the elements of the visibility map array with corresponding elements of the Planetary Transit Function array.

To be able to do that, we require that the two array have the same number of elements. However, the Planetary Transit Function is piecewise continuous by definition. So we first have to convert that into an array by discretizing the continuous timesteps into that of 6-minute cadence so that it would match up with the timesteps cadence of the visibility map (see Figure 5.1). This can be done using the numpy vectorize function.

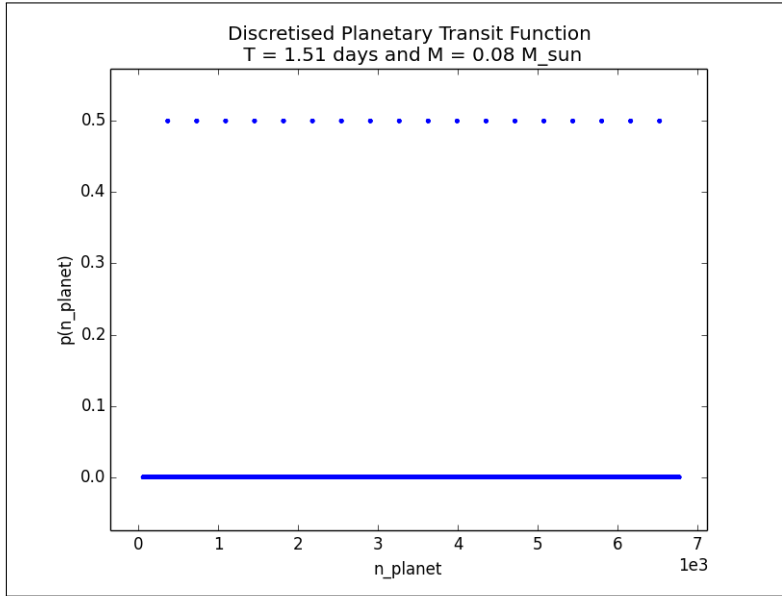


Figure 5.1: Discrete Planetary Transit Function.

Again, notice that the Planetary Transit Function does not start at $n_{planet} = 0$ with the random orbit location generator activated. Also, the values in the x-axis here are completely arbitrary, we can replace them with any value (e.g. the SkyHopper observation timesteps) and it will not change anything. The values in the x-axis here is chosen for plotting and illustration purposes only.

Now after all that has been done, recall that the visibility map is a binary, where 1 mean the selected coordinate is visible and 0 means otherwise. Then we also have the Planetary Transit Function defined by 0.5 during transit and 0 for non-transit. So by convolution, we get a new array

of the same shape with values of 0.5 and 0. Let's call this new array "Potential Transit Detections" and we will see why it is name that way later.

Elements in this array with a value of 0.5 tells us that the visibility = 1 timesteps of SkyHopper coincides with a planetary transit timesteps, which also means that SkyHopper is observing a potential transit timestep. Whereas 0 means either no transit or selected location is not visible by SkyHopper. This can be visualised by plotting the "Potential Transit Detections" array against the respective timesteps as shown below.

Since transits are represented by the 0.5 mark in the y-axis, we can count the number of intrinsic transit that the planet has and also the number of potential transits detected by SkyHopper by counting the number of "points" in the 0.5 regime of Figure 5.1 and 5.2 respectively. Intrinsic transit here refers the the actual total number of transits that the has occurred. You might also notice why is the word "points" in inverted commas - refer to Figure 5.3 to see why.

Now we direct our attention to the pattern of Figure 5.2 which looks very similar to that of Figure 5.1. However, there are missing "points" (i.e. empty gaps) in Figure 5.2. These gaps indicate the transits missed by SkyHopper and we can use that later to obtain an approximate transit detection efficiency of SkyHopper.

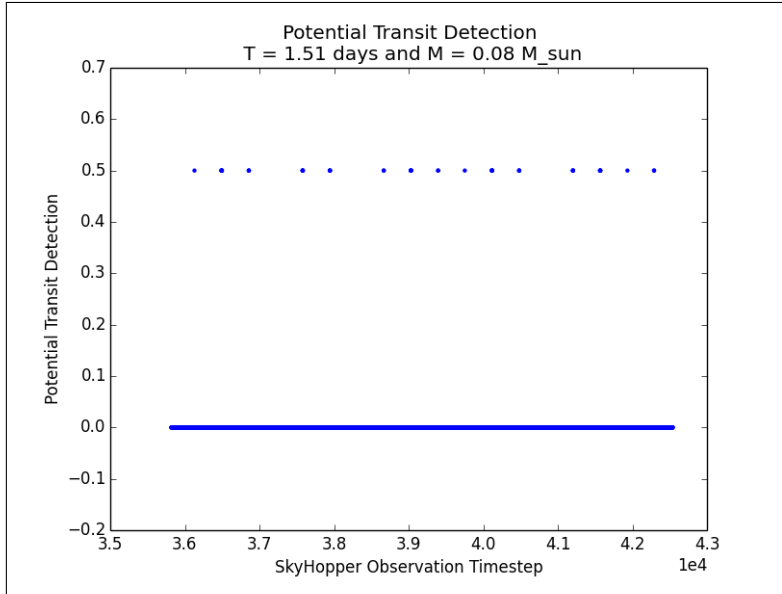


Figure 5.2: As mentioned, the "points" at $y = 0.5$ represent potential transit detection. However, they are not exactly points. See the next figure for explanation.

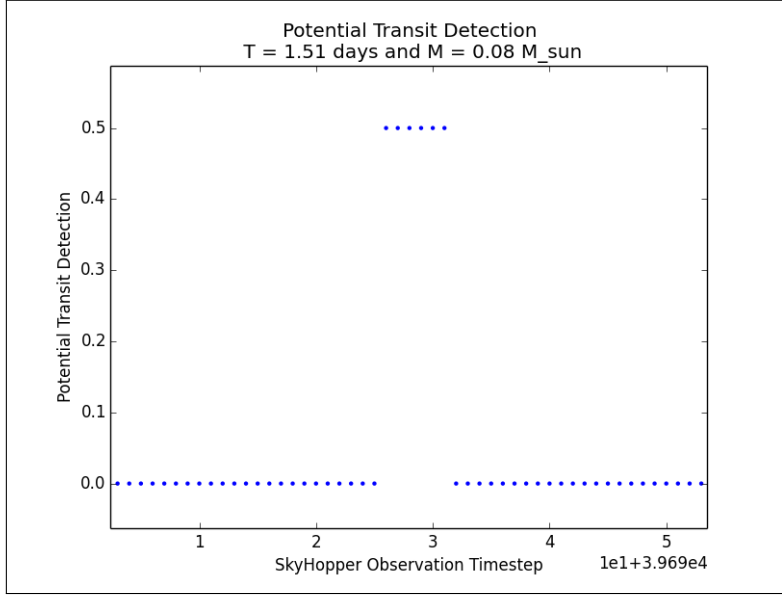


Figure 5.3: This figure shows what happens if we zoom in to one of the “points” at $y = 0.5$. They are not actually a single point. The number of adjacent points here actually tells us the exact number of potential transit timesteps SkyHopper has observed.

5.2 Detailed Visualisation of Potential Transit Detection

This section essentially has nothing to do with the simulation results, but it does show some pretty colour coded plots which visualises exoplanet transit detection.

An extra array is required to produce the color coded plot. This array is similar to the “Potential Transit Detections” array, but instead of multiplying the elements in the visibility map and transit function, we will add them here. By adding rather than multiplying, we will obtain more information from the resulting array where:

- value = 1.5 : transit is detected
- value = 1.0 : non-transit is detected
- value = 0.5 : transit is missed
- value = 0.0 : non-transit is missed

Using the information of the two generated arrays, we can filter out specific data points to generate a colour coded plot for visualisation. The data points on this plot is only for when SkyHopper visibility is 1 and it is essentially plotting SkyHopper visibility and Planetary Transit Function on the same graph (see Figure 5.4).

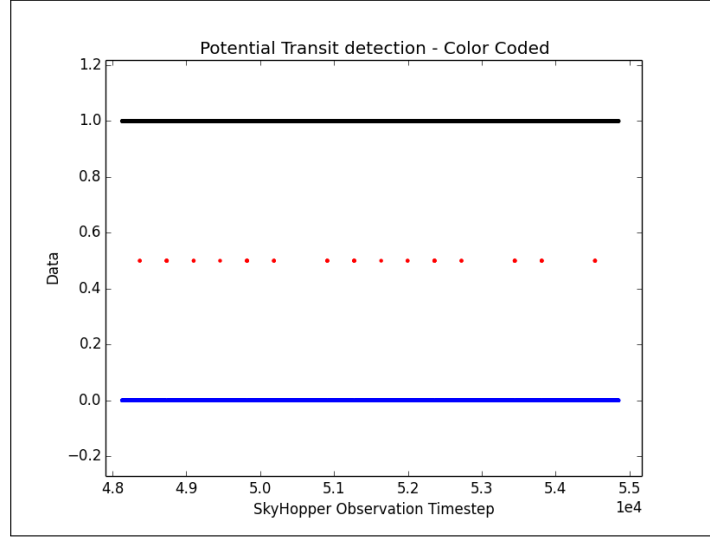


Figure 5.4: Colour coded SkyHopper visibility = 1 and Transit Function Plot

The special thing about this plot is that the colors on the plot represents different events:

- Red: Transit detected
- Black: Non transit detected
- Blue: Non-transit orbital timesteps

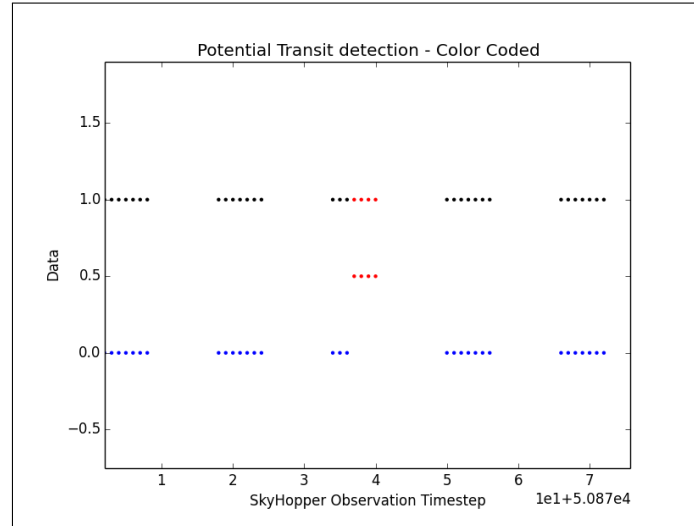


Figure 5.5: Zoomed in version of Figure 5.4 to give a more detailed perspective on what is happening for each coded color.

6 TRANSIT DETECTION EFFICIENCY

6.1 Exoplanet Transit Detection Condition

We now come back to the question as to why in Section 5 we kept including the word “Potential” before exoplanet detection. The idea to that is, even though SkyHopper may have detected a luminosity drop, we cannot be certain that it has detected a transit. The luminosity drop detected must last long enough to be identified as a transit. Another thing is, we have to ensure that we can see the luminosity dip or rise at the beginning or at the end of the planetary transit respectively. Otherwise, if SkyHopper detects only something of low luminosity during that visibility period, it may look like a planetary transit or a luminosity drop, but it could also possibly be a scaling error or some other instrumental error.

In this simulation, the definition of a detected transit is that during each visible period of SkyHopper, there must be at least 2 timesteps each of transits and non-transit detected. So something like 5 transit steps and 3 non-transit steps (Figure 6.1a) would pass the filter condition while something like 7 transit steps and 1 non-transit step (Figure 6.1b) would not.

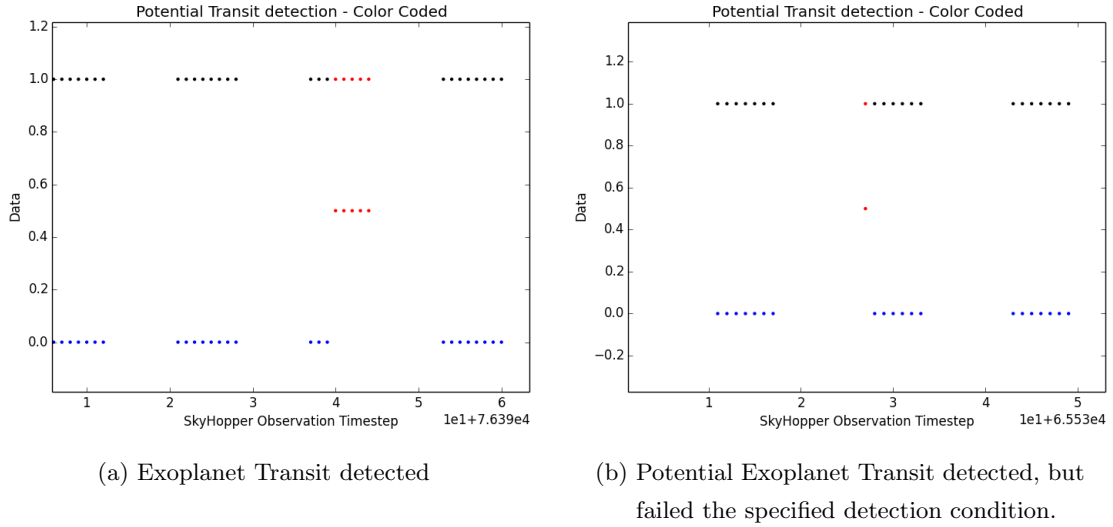


Figure 6.1

6.2 Transit Detection Efficiency

Recall that from Figure 5.1, we could count the number of intrinsic transits made by the planet around the star. And now that we have the number of detected transits which passed the selection condition, k_{pass} , we can easily calculate SkyHopper’s efficiency of exoplanet by taking the ratio of k_{pass} to the number of intrinsic transits, k_{tot} (see Equation 9 next page).

SkyHopper Transit Detection Efficiency, ϵ :

$$\epsilon = \frac{k_{detected}}{k_{tot}} \quad (9)$$

NB: k_{tot} is not necessarily an integer. There could be 3.5 total intrinsic transits that have occurred over the observation period.

This efficiency is arguably the most crucial result from the Monte Carlo simulation of exoplanet transit detection. Of course, the higher efficiency the better, but we are not always that lucky to catch the maximal amount transits with SkyHopper. Therefore, we have to perform the simulation many times by looping the algorithm to look at the general efficiency of SkyHopper's ability to detect exoplanet transits (see Section 109238 for efficiency distribution plot).

7 SIMULATION OUTPUT

7.1 Sample Monte Carlo Simulation

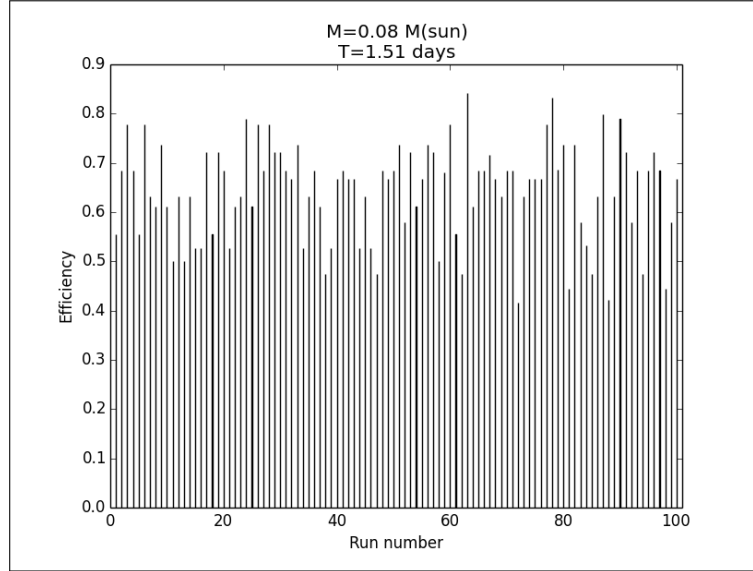


Figure 7.1: Transit detection efficiency distribution for 100 runs

7.2 Output Plots for Individual Run

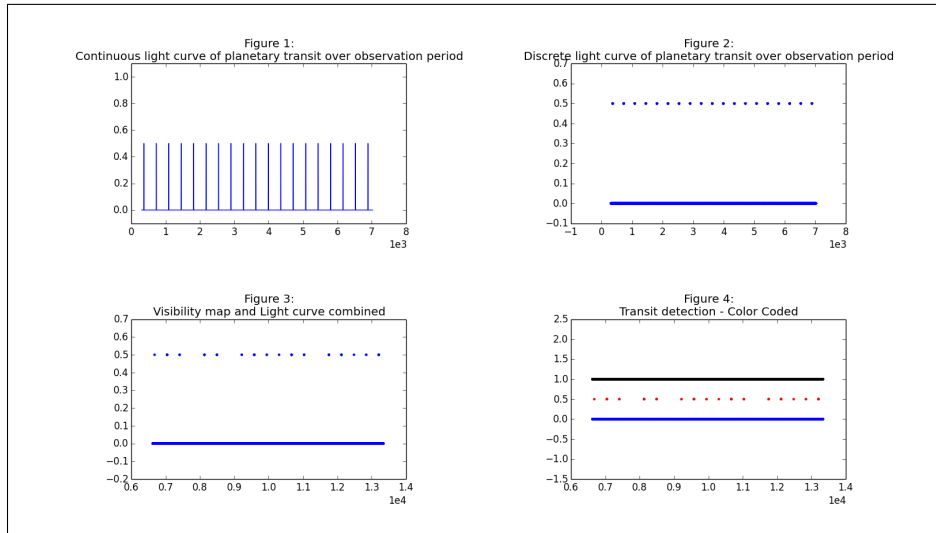


Figure 7.2

7.3 General Information Output

This output consists of the general information of the simulation. The “General Details” section provide information on the parameters which we are testing for the particular simulation.

```
*** COMMENTS ON FIGURE 3 *****
if y = 0.5:  Transit detected
if y = 0:   Selected region not visible to SkyHopper/No transit

*** COMMENTS ON FIGURE 4 ***
Red:  Transit detected
Blue:  Non-transit orbital timesteps
Black: SkyHopper with visibility = 1 but no transit detection
*NB: SkyHopper timesteps with visibility = 0 are not shown in this plot

*** TRANSIT DETECTION CONDITION ***
Condition of detection: Must have at least 2 transit and 2 non-transit
steps in one SkyHopper visibility period

*** GENERAL DETAILS ***
Stellar mass = 0.080 M(sun)
Orbital period = 2174.400 min
Semi-major axis = 1.663e+09 meters
Transit time in minutes = 38.5 min

*** PLANETARY TRANSIT DETAILS ***
No.  of timesteps in 1 planet orbit = 362.40
No.  of timesteps in 1 planet transit = 6.42
No.  of timesteps for non-transit time = 355.98
```

7.4 Information for Individual Runs

As mentioned before, we have to loop over the whole algorithm a number of times to obtain an average SkyHopper transit detection efficiency. Since every run have different initial conditions (i.e. SkyHopper observation start time and planet orbital location at the start of observation), they will give different detection efficiencies through the simulation. Hence for every individual run, the script will write out both the initial conditions and its outcome.

The output shown below is a sample of the information recorded from the simulation:

*** RUN NUMBER 1 ***

Observation time = 4 week(s)

*** VISIBILITY MAP DETAILS ***

Visibility map timestep cadence = 6.0 min

Visibility map starting timestep (random) = 37116

Ending timestep = 43837

*** COORDINATES ***

[Ra,Dec] = [0,0]

*** ORBIT START LOCATION ***

Orbit start point (random) = step 223.1617186659 of 362.400

Planet orbit location = 0.62

*** INTRINSIC TRANSIT COUNTER ***

Number of partial transits = 0

Number of full transits = 19

Number of steps in first partial transit = 0

Number of steps in last partial transit = 0

Fraction of first partial transit = 0.0000

Fraction of last partial transit = 0.0000

Total intrinsic transit = 19.0000

*** PROBABILITY OF TRANSIT DETECTION ***

Number of transit timesteps over observation period = 121

Number of coinciding timesteps = 58

Transit visibility efficiency = 0.4793

*** TRANSIT DETECTION ***

Number of Potential Transits Detected = 16 out of 19 intrinsic transit(s)

Number of Actual Transits Detected = 13 out of 16 possible candidate(s)

Detection efficiency = 0.684

The run number is indicated at the very beginning before recording the information to allow data tracing if interesting results (e.g. unusually high efficiency) are produced for that specific run.

7.5 Transit Steps Detected per Orbit

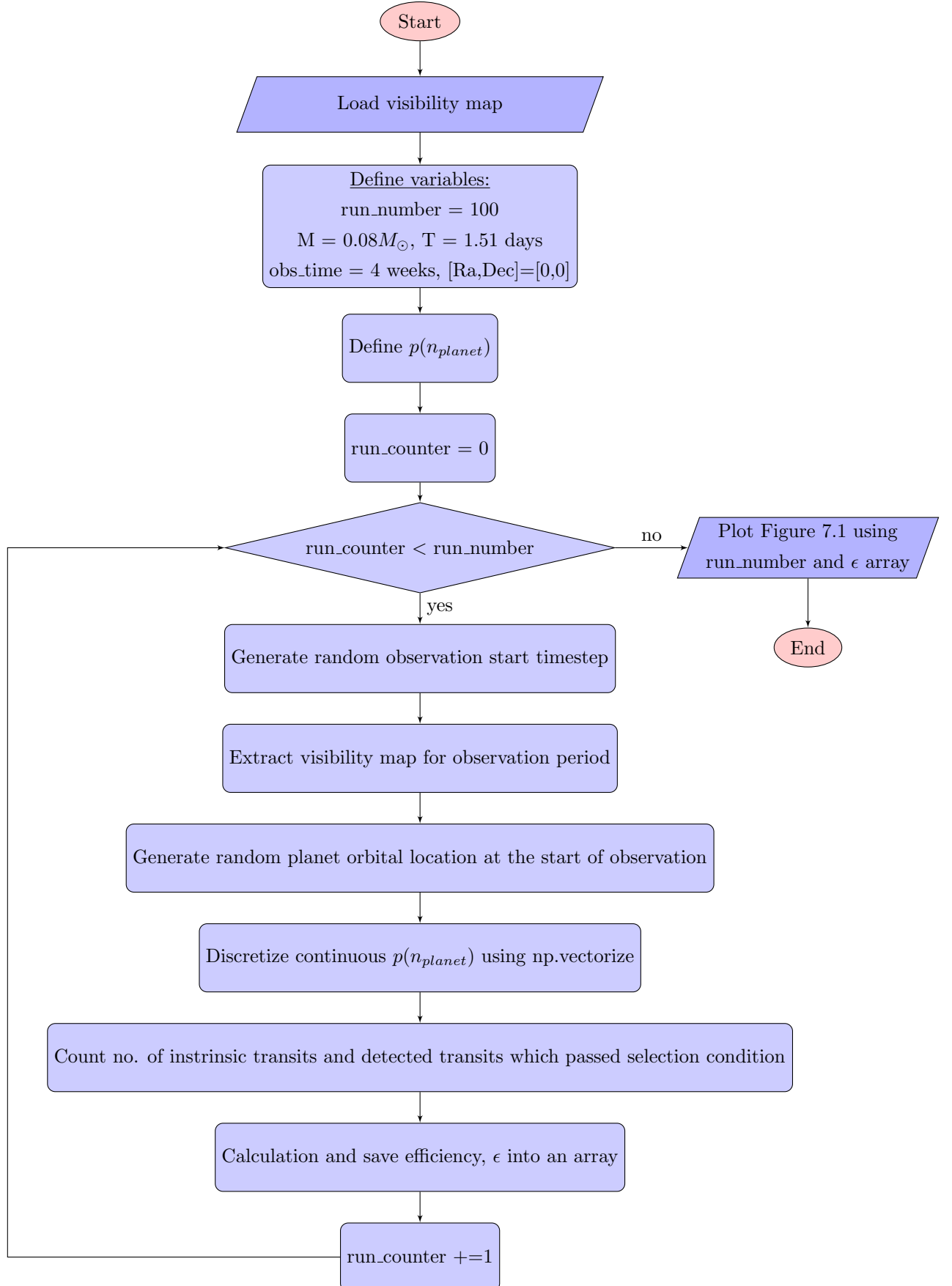
This information records the number of transit steps detected by SkyHopper in each of its orbit for each individual run in the Monte Carlo simulation. However, it only records those which passed the selection condition specified in Section 6.1. Here is an example:

```
***RUN NUMBER 1***
```

```
Transit steps detected in orbit 52 = 5.00  
Transit steps detected in orbit 75 = 3.00  
Transit steps detected in orbit 120 = 4.00  
Transit steps detected in orbit 143 = 4.00  
Transit steps detected in orbit 188 = 2.00  
Transit steps detected in orbit 234 = 2.00  
Transit steps detected in orbit 279 = 6.00  
Transit steps detected in orbit 302 = 3.00  
Transit steps detected in orbit 347 = 3.00  
Transit steps detected in orbit 369 = 2.00  
Transit steps detected in orbit 370 = 3.00  
Transit steps detected in orbit 415 = 2.00
```

The orbit numbers here represent the n-th SkyHopper orbit since the start of observation. To be clear, each SkyHopper lasts for 96 minutes (16 timesteps of 6-minute cadence).

8 SUMMARY OF SIMULATION (FLOWCHART)



9 WHAT'S NEXT?

All in all, although the simulation is producing decent results that are within expectations, there is still room for improvements.

A few suggestions are that the simulation can be modified to:

- loop over all-sky as it is only looking at one specific [Ra,Dec] at the moment
- have a more well-defined scaling in terms of the power of 2 rather than 0.0, 0.5, 1.0, and 1.5
- include flags as to when and where do we want to produce figures rather than to produce all the figures. Otherwise for a large number of runs, it may take a very long time to compute which is impractical.
- take into account instrumental noise
- fit with actual light curve using box least squares (i.e. least square fitting)

Also, recall that assumptions have been made in this simulation:

- All observed orbital plane is parallel to line of sight
- Instantaneous flux change at the start and end of transits

Hence we can also start taking into account the inclination of the orbital plane and also the gradual flux change at the start and end of transit.

References

- [1] SkyHopper - For Scientist. <https://skyhopper.research.unimelb.edu.au/for-scientists/>. Accessed: 7 July 2018.
- [2] NASA's Jet Propulsion Laboratory (California Institute of Technology). SEARCHING FOR SHADOWS: Transit. <https://exoplanets.nasa.gov/5-ways-to-find-a-planet/#/2>. Accessed: 20 July 2018.
- [3] Vera Maria Passegger. The Paradox of Overfitting. Master's thesis, Universität Wein, Universitätsring 1, 1010 Wien, Austria, 2013.